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Topological Uniqueness of the Nash Equilibrium for Non-cooperative Routing

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Abstract

We consider the problem of selfish routing in a congested network with finitely-many users, where each user wishes to minimize the cost of its own flow. Flows may be split over different routes. The cost function is additive, with user-specific link costs that may depend on the user's own flow as well as on the total link flow. Our main interest here is network topologies that ensure the uniqueness of the Nash equilibrium under mild convexity assumptions on the link costs. We characterize the class of two-terminal network topologies for which this uniqueness property holds, and show that this class coincides with the set of *nearly parallel networks* that was recently introduced by Milchtaich [20] in the context of Wardrop equilibrium analysis. We further show how our results imply and extend known results on uniqueness of the Wardrop equilibrium in multi-class networks.

Key Words: selfish routing, congested networks, Nash equilibrium
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